

THE GIANT STEPS

- Identify and address large-scale societal issues through out-of-the-box, tech-based solutions
- Nurture young talent, empower them to take on ambitious, problem-solving initiatives
- Prioritise design, development and local component manufacturing to increase value addition

Game-changing systems: Today, in a city like Chennai, it takes over an hour to travel about 15 km between home and office. Wasting over two hours on a daily commute cannot be our future. It is likely to get worse with increasing urban migration and vehicle ownership. Suppose we design a transport system that could make this journey possible in 20 minutes, even in peak time, while addressing some additional issues—be the most energy-efficient transport system (ruling out ideas like flying), and use only green electricity; scale for tomorrow's peak traffic and provide a very comfortable ride (say with air-conditioning); be deployable in existing cities with all their constraints; and have a total cost (including depreciation, interest and operating expenditure) less than that of travelling by a two-wheeler today. To conceive, build and operationalise such systems would be game-changing, not just for India, but also for similarly crowded cities elsewhere. This will require multiple technologies and push us to be a world leader.

Virtual experts: Let us take another Indian problem. About 70 per cent of Indian households have monthly incomes less than ₹15,000. When a woman from such a household wants to start a nano-enterprise and needs a mere Rs 1 lakh loan, the first choice is invariably to go to informal sources, where annual interest rates can be as high as 36-60 per cent. Over the last 15 years, microfinance and NBFCs have emerged to provide such loans at about 25 per cent interest rates; their cost of funds, cost of risks, and cost of operations are such that bringing the rates much lower is a serious challenge. There are experts who could visit the woman at her home, talk with her for about 10 minutes, and pose the next question listening carefully to each answer, so as to assess the risk associated with the desired loan accurately. The problem is that such experts do not scale and sending an inexperienced person to do the job increases the risks considerably.

It is here that new technologies can solve the problem non-linearly. Fortunately, mobile phones and WhatsApp video calls have reached even such low-income homes. Suppose we leverage technologies like AI, video and speech processing to send a pleasant-sounding Virtual Expert, who speaks in the local language and dialect with the woman. It would examine the words, expressions and tone of each response and at the same time search for any information about the woman and her financial transactions to pose the next question. One would not only be able to determine risks associated with the business and the loan but could also use similar calls to carry out collections. The costs associated would thus go down considerably, and loans at 15 per cent interest rates would become viable. There is no reason why India cannot build such a technology, use it widely to perfect it and take it to the world.

There are indeed many such large avenues to become the technology leader in the world. We can get there only if we think afresh and have a strong will. ■

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